Network Analysis Report

1. **Number of Devices Discovered:**

8 active hosts discovered by the nmap scan on the local subnet 192.168.100.0/24

- *Active devices found:*

✓ A Huawei Technologies router/gateway (192.168.100.1).
✓ A Dell device (192.168.100.134).
✓ Mobile devices from Samsung, vivo, and LG.
✓ An unknown device with MAC address `B6:64:08:F5:53:64` (192.168.100.181).

- *Key findings:* Device with 192.168.100.166 IP heavily involves in the capture traffic, also mentioned in the nmap scan as an active host.

2. Common Protocols Observed:

- *DNS (Domian Name System):* Primarily used for resolving Mozilla/Firefox domains `(e.g., content-signature-2.cdn.mozilla.net, ads.mozilla.org, push.services.mozilla.com).`
- *TCP (Transmission Control Protocol):* For establishing connections, this protocol is considered as a foundational transport protocol.
- *TLS v1.2 (Transport Layer Security):* Provides communication security, encrypted traffic handling secure web browsing and server authentication.
- *HTTP (HyperText Transfer Protocol):* Unencrypted web traffic like, OSCP request/response and an HTTP GET request.
- *OCSP (Online Certificate Status Protocol):* It is used to check the void status of SSL/TLS certificate (observed in the HTTP capture window).

3. Key Observations from the Traffic Capture

1. *The user was using Firefox to browse the internet*

Almost all the network traffic we captured was related to Firefox. We saw the computer asking for the web addresses of Mozilla servers, starting a secure connection to a Mozilla website, and checking a digital security certificate. This means the person using this computer was actively browsing the web with Firefox.

2. *A connection attempt got blocked or failed*

We saw one computer trying to start a connection with a remote server (by sending a "SYN" packet). However, the server didn't reply to say "I'm ready." Instead, a different computer sent a "Reset" command (RST), which means "I'm closing this connection." This shows that a connection attempt failed, either because the server refused it or there was a network issue preventing the connection from being made.

3. *Some data was lost while recording the traffic*

In the capture, there is a message that says "previous segment not captured." In simple terms, this means Wireshark missed catching a piece of data that was sent over the network. This usually happens because the computer was working too hard to catch all the data at once, or the recording buffer got full. Because of this missing data, we can't see the full picture of that specific web response.

4. *The web traffic was safely encrypted*

Most of the actual web traffic (like the data being sent and received) was heavily encrypted using a security standard called TLS 1.2. It also included checks to verify website identities (OCSP). This is a good thing, it means the user was visiting websites securely, and anyone snooping on the network wouldn't be able to easily read the information being sent back and forth.